Faculdade de Ciências da Universidade de Lisboa

Department of Mathematics

3rd Year – 2nd Semester – 2025/2026
Course Unit: Statistical Consulting/Project (22718)
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Statistical Consulting Project

Analysis of Average Daily Weight Gain in Charolais Calves

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Executive Summary

At a 5% significance level, there is **no** statistical evidence that the mean of the Average Daily Weight Gain (ADG) is significantly greater than the sample median (1.745).

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1 Introduction

This project analyzes the ADG of a sample drawn from a population of charolais calves to an experimental diet containing vitamins additives, niacin, and antibiotics. The objective is to test wheter the mean value of the ADG is significantly greater than the sample median. We will divide this project into two cases, with Case 1 being the full sample and Case 2 the sample without outliers.

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2 Main Results and Conclusions

In the analysis of the full sample (Case 1), the presence of atypical values was detected. These atypical values (outliers) led to the rejection of the normality assumption, compromising the project's main objective. For this case after perfoming the appropriate statistical tests, it was concluded that there was no statistical evidence that the population median was greater than the sample median.

In the anlysis of the sample without outliers (Case 2), conducted to avoid distortion of the results, the assumption of normality was not rejected. Consequently, the mean value of the ADG showed no statistical evidence of being greater than the sample median.

3 Results

3.1 Case 1: Full Sample ($n = 42$)

The following table presents the main descriptive statistics for the full sample, alongside the results of the perfomed statistical tests. It can be observed that the presence of the 1.07 kg (outlier) caused a strong negative skewness of -1.850, which is visible both the histogram and the box plot. This strong skweness in the data distribuition led to the rejection of the the normality assumption, proceeding with the use of the Sign Test, which yielded a p-vlue of 0.561. Therefore, there is no statistical evidence to claim that the population median of the ADG is greater than the sample median (1.745 kg).

Statistic	**Value**
Sample average	1.746 kg
Sample median	1.745 kg
Sample standard deviation	0.149 kg
Coefficient of variation	8.55%
Skewness	-1.850
Identified outliers (Box plot)	1.07 kg & 2.06 kg
W_{obs}	0.819
p-value (Shapiro-Wilk)	< 0.05
S_{obs}	21
p-value (Sign Test)	0.561

Table 1: Full sample statistics

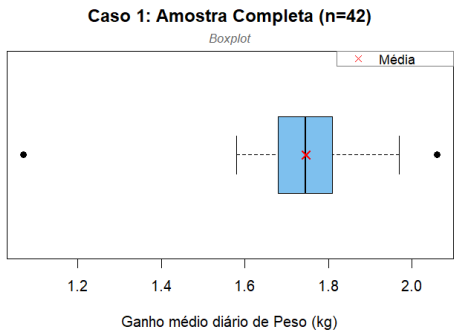

Figure 1: Box plot (Case 1)

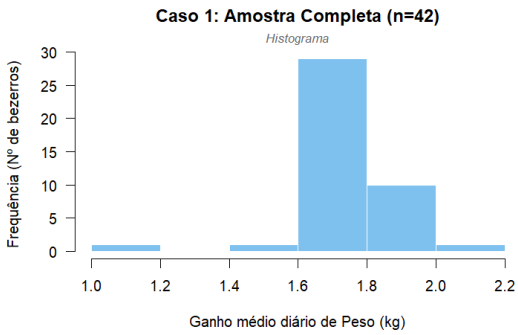

Figure 2: Histogram (Case 1)

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3.2 Case 2: Sample without Outliers($n = 40$)

The following table presents the main descriptive statistics for the sample without outliers, alongside the results of the performed statistical tests. By removing the outliers, it can be observed that the sample mean increased to 1.756 kg and the skewness decreased to 0.557, resulting in a more symmetrical data distribuition, as illustrated in the box plot and histogram. This resulted in the validation of the normaility assumption.

The application of the Student's t-test yielded a p-value of 0.246. Therefore, there is no statistical evidence that the mean value of the ADG is greater than the sample median (1.745 kg).

Statistic	**Value**
Sample mean	1.756 kg
Sample median	1.745 kg
Sample standard deviation	0.095 kg
Coefficient of variation	5.44%
Skewness	0.557
W_{obs}	0.955
p-value (Shapiro-Wilk)	0.113
T_{obs}	0.695
p-value (Student's t-test)	0.246

Table 2: Statistics of sample without outliers

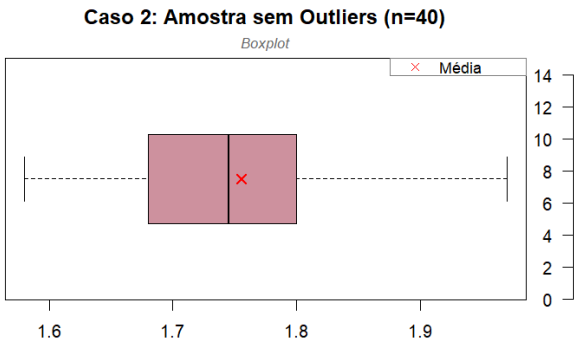

Figure 3: *Box plot* (Case 2)

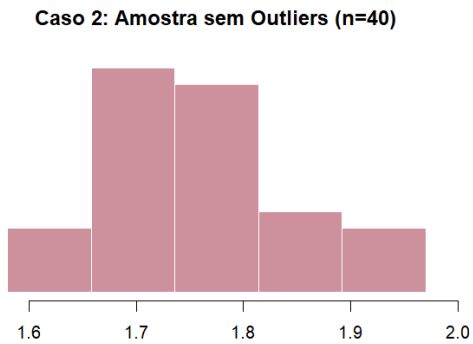

Figure 4: Histogram (Case 2)

4 Methodology

Throughout this project, we assume:

- Let X be the random variable representing the sample of charolais calves fed a diet containing vitamin additives, niacin, and antibiotics;
- Significance level of $\alpha = 0.05$;
- Rejection rule: p-value $< \alpha$;

The validation of the normality assumption in both cases was performed using the **Shapiro-Wilk test**, structured as follows:

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$$\mathrm{H}_0 : X \frown \mathcal{N}(\mu, \sigma) \quad vs. \quad H_1 : X \not\frown \mathcal{N}(\mu, \sigma) \qquad E.T : W = \frac{\left(\sum_{i=1}^{n} a_i x_{(i)}\right)^2}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

Based on the results of the Shapiro-Wilk test, the hypothesis testing formulation differed between the two cases.

4.1 Case 1: Full Sample ($n = 42$)

Since we rejected H_0 in the Shapiro-Wilk test, we must follow a non-parametric approach. However, non-parametric tests do not assess the mean value (μ); therefore, we will use the closest parameter of central tendency, which is the population median ($\chi_{1/2}$).

For a one-sample median test, we can only use the Wilcoxon Signed-Rank Test and the Sign Test. However, the Wilcoxon test can only be applied under the assumption that the distribution is symmetric. Therefore, we must use the **Sign Test**, structured as follows:

$$H_0 : \chi_{1/2} \leq 1.745 \text{ kg} \quad vs. \quad H_1 : \chi_{1/2} > 1.745 \text{ kg} \qquad E.T : S = \sum_{i=1}^{n} I(X_i > \chi_{1/2})$$

Under H_0, S follows a Binomial distribution $B(n^*, 0.5)$, where n^* is the number of observations with values greater than the median (excluding ties with the sample median).

Note: Since $n > 30$, we could have used the Central Limit Theorem (CLT) to assume that the sample mean is approximately Normally distributed, despite having rejected H_0 in the Shapiro-Wilk test. However, we decided to adopt a more conservative approach and perform non-parametric tests.

4.2 Case 2: Sample without Outliers ($n = 40$)

By removing the outliers, we failed to reject H_0 in the Shapiro-Wilk test, and therefore we must follow a parametric approach. Since in this case the normality assumption is validated, the variable is quantitative and continuous, the population variance is unknown, and we are testing against a reference value, a **One-Sample Student's t-test** must be performed, structured as follows:

$$H_0 : \mu \leq 1.745 \text{ kg} \quad vs. \quad H_1 : \mu > 1.745 \text{ kg} \qquad E.T : T = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}}$$

Under H_0, T follows a Student's t-distribution with $n - 1$ degrees of freedom.

5 Contextualization

The charolais cattle breed is recognized in the livestock industry for its biological and genetic predisposition to achieve higher weight gains than other breeds. This characteristic explains the high ADG recorded in the sample, as the experimental diet sought to maximize the potential of the charolais calves.

Due to this weight-gaining capacity, charolais cattle are frequently crossbred with other breeds so that the offspring can inherit these genetic traits.

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6 Final Considerations

It is essential to critically evaluate the study design. Testing whether the mean value is significantly greater than the sample median primarily allows us to infer the presence of positive skewness in the data distribution.

Firstly, increasing the sample size is recommended to enhance the representativeness of the population.

Additionally, if the goal is to determine whether the experimental diet outperforms the standard diet, it is highly suggested to include a group from the same population receiving the standard diet. This approach enables a direct comparison with the experimental diet and helps assess whether the financial investment in the experimental diet is justifiable.

Notice:

- In this report, we utilized Artificial Intelligence (AI - Google Gemini) to support and validate our ideas. The core reasoning and critical thinking are entirely our own, with the AI serving strictly as a consultation tool to verify our logic.
- This report was written in accordance with the guidelines described in Ehrenberg's article.
- The results, values, and charts presented in this report were generated using the R statistical software.
- The Q-Q plots can be found in the appendix.

References

[1] Ehrenberg, A. S. C. (1982). *Writing Technical Papers or Reports*. The American Statistician, 36(4), 326-329. American Statistical Association. (Accessed on March 6, 2026).

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[3] GardenTime (n.d.). *Raça de vacas Charolês: descrição e características*. Available at: `https://gardentime-pt.decorexpro.com/hozyajstvo/zhivotnovodstvo/sharole-poroda-korov-opisanie` (Accessed on March 7, 2026).

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[5] Severino, E. (2025/2026). *Sebenta de Estatística Não Paramétrica*. Course support material. Faculdade de Ciências da Universidade de Lisboa. Available at: `https://moodle.ciencias.ulisboa.pt/pluginfile.php/612491/mod_resource/content/1/Estat%C3%ADstica%20N%C3%A3o%20Param%C3%A9trica_Sebenta.pdf` (Accessed on March 10, 2026).

[6] Domingues, T. D. (2024/2025). *Estatística Paramétrica (Cód. 22706): Estimação (Parte I)* [Presentation slides]. Departamento de Estatística e Investigação Operacional, Faculdade de Ciências da Universidade de Lisboa. (Accessed on March 12, 2026).

A Appendix

This script contains all the R code developed for the descriptive statistics, data visualization, and execution of the hypothesis tests presented in this report.

```r
library(moments)

# 1 Projeto  - Consultoria
# Faculdade de Ci ncias da Universidade de Lisboa
# Departamento de Ci ncias Matem ticas
# 3.   Ano - 2.   Semestre - 2025/2026
# Projeto realizado por: Ricardo Morais(62802), Martim Diniz(62049)
# Projeto realizado no  mbito  da cadeira de Consultoria em Estat stica/Projeto(c digo: 22718),
# Docente respons vel: Maria Fernanda Diamantino
# Docente auxiliar: Ana Catarina Fernandes



parmfrow = c(3,3)
ganho <- c(
  1.07, 1.66, 1.71, 1.75, 1.78, 1.89,
  1.58, 1.66, 1.71, 1.75, 1.79, 1.89,
  1.63, 1.67, 1.72, 1.76, 1.79, 1.93,
  1.64, 1.68, 1.72, 1.77, 1.81, 1.94,
  1.64, 1.68, 1.73, 1.77, 1.83, 1.94,
  1.66, 1.69, 1.74, 1.77, 1.84, 1.97,
  1.66, 1.69, 1.74, 1.78, 1.86, 2.06
)

mean(ganho)
median(ganho)
sd(ganho)
skewness(ganho)
cv1 <- (sd(ganho) / mean(ganho)) * 100
cv1

# par(mfrow = c(2,2))

# Boxplot Caso 1
boxplot(ganho, horizontal = TRUE, col = "skyblue2",
        main = "Caso 1: Amostra Completa (n=42)",
        xlab = "Ganho m dio di rio de Peso (kg)",
        pch = 16, medcol = "black", medlwd = 2)
mtext("Boxplot", side = 3, line = 0.3, cex = 0.85, col = "gray40", font = 3)
points(mean(ganho), 1, pch = 4, col = "red", lwd = 2, cex = 1.2)
legend("topright", legend = "M dia", pch = 4, col = "red", bty = "o", box.col = "gray50", bg = "white", cex = 0.9)


# Histograma Caso 1
hist(ganho, breaks = "sturges",
     col = "skyblue2", border = "white",
     main = "Caso 1: Amostra Completa (n=42)",
     xlab = "Ganho m dio di rio de Peso (kg)", ylab = "Frequ ncia (N de bezerros)", las = 1)
mtext("Histograma", side = 3, line = 0.3, cex = 0.85, col = "gray40", font = 3)

length(ganho)

# H0 : X     N(  ,   ) vs. H1 : X       N(  ,   )
shapiro.test(ganho)
```


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```r
# p-value < alpha, logo rejeitamos a hip tese da normalidade
# QQ-Plot Caso 1
qqnorm(ganho,
       main = "Q-Q Plot: Caso 1 (n=42)",
       xlab = "Quantis Te ricos (Normal)",
       ylab = "Quantis Emp ricos (Amostra)",
       pch = 16, col = "black", cex = 1.2, las = 1)
qqline(ganho, col = "red", lwd = 2)
mtext("An lise de Normalidade", side = 3, line = 0.3, cex = 0.8, col = "gray40", font = 3)



mediana_ref <- median(ganho)
ganho_valido <- ganho[ganho != mediana_ref]
sucessos <- sum(ganho_valido > mediana_ref)
n_total <- length(ganho_valido)
teste_sinais <- binom.test(x = sucessos,
                           n = n_total,
                           p = 0.5,
                           alternative = "greater")
teste_sinais


# --- Caso 2 ---

ganho2 <- c(
  1.66, 1.71, 1.75, 1.78, 1.89,
  1.58, 1.66, 1.71, 1.75, 1.79, 1.89,
  1.63, 1.67, 1.72, 1.76, 1.79, 1.93,
  1.64, 1.68, 1.72, 1.77, 1.81, 1.94,
  1.64, 1.68, 1.73, 1.77, 1.83, 1.94,
  1.66, 1.69, 1.74, 1.77, 1.84, 1.97,
  1.66, 1.69, 1.74, 1.78, 1.86
)

mean(ganho2)
median(ganho2)
sd(ganho2)
cv2 <- (sd(ganho2) / mean(ganho2)) * 100
cv2

# Boxplot Caso 2
boxplot(ganho2, horizontal = TRUE, col = "pink3",
        main = "Caso 2: Amostra sem Outliers (n=40)",
        xlab = "Ganho m dio di rio de Peso (kg)",
        pch = 16, medcol = "black", medlwd = 2)
mtext("Boxplot", side = 3, line = 0.3, cex = 0.85, col = "gray40", font = 3)
points(mean(ganho2), 1, pch = 4, col = "red", lwd = 2, cex = 1.2)
legend("topright", legend = "M dia", pch = 4, col = "red", bty = "o", box.col = "gray50", bg = "white", cex = 0.9)


cortes <- seq(min(ganho2), max(ganho2), length.out = 6)
# Histograma Caso 2
hist(ganho2, breaks = cortes,
     col = "pink3", border = "white",
     main = "Caso 2: Amostra sem Outliers (n=40)",
     xlab = "Ganho m dio di rio de Peso (kg)", ylab = "Frequ ncia (N de Bezerros)",
     las = 1, xlim = c(1.55, 2.0))

```


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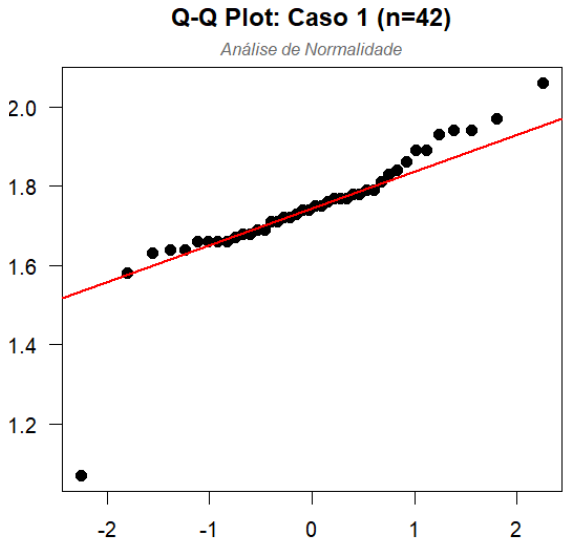
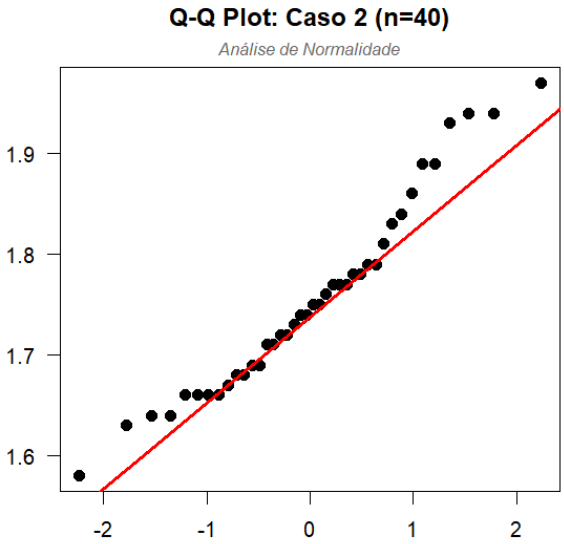
```r

# Coeficiente de assimetria: g1 = (1/n * sum( x i x )^3) /    ((1/n * sum( x i x )^2)^3)
skewness(ganho2)
length(ganho2)

# H0 : X     N(  ,   ) vs. H1 : X       N(  ,   )
shapiro.test(ganho2)
# p-value = 0.1126 > 0.05 = alpha
# QQ-Plot Caso 2
qqnorm(ganho2,
       main = "Q-Q Plot: Caso 2 (n=40)",
       xlab = "Quantis Te ricos (Normal)",
       ylab = "Quantis Emp ricos (Amostra)",
       pch = 16, col = "black", cex = 1.2, las = 1)
qqline(ganho2, col = "red", lwd = 3)
mtext("An lise de Normalidade", side = 3, line = 0.3, cex = 0.8, col = "gray40", font = 3)

t.test(ganho2, mu = 1.745, alternative = "greater", conf.level = 0.95)
```





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